

LLM for Teens

By Hamid Sadeghi

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■ Agentic AI

Agentic AI is an AI system that can act on its own to achieve a goal, not just respond to one prompt.

Instead of answering a single question and stopping, an agentic system plans, takes actions, checks results, and keeps going until the task is done. It uses tools — search, APIs, code, databases — and decides which ones to use at each step.

■ Tools

An LLM by itself can only generate text. It can't check the weather, search a database, or send an email.

A tool is a function the LLM is allowed to call to do something outside itself.

The LLM doesn't run the tool. It just says which tool to call and with what inputs. Your program runs it, then sends the result back to the LLM.

Without tools, the LLM can only talk. With tools, it can act.

MCP is the standard way to publish and share those tools across different AI apps. Tools are abilities we give to an LLM. Without tools, the LLM can only generate text. MCP is the standard way to publish those tools, so any app that supports MCP can use them, no matter which LLM runs inside.

■ MCP

MCP (Model Context Protocol) is an open standard that lets AI models connect to external tools and data sources in one consistent way. Think of it as a USB-C port for AI.

■ Where to Run a MCP Server

- Your computer
- Kaggle
- Google Colab
- GitHub Codespaces
- Hugging Face Spaces
- Render
- Railway
- Fly.io
- Google Cloud Run
- Azure

Assignment

- In this assignment, you will build a small MCP server with one tool and connect it to a Python client. The goal is to see how a model discovers and calls external tools through a standard protocol.
- Dataset
 - Kaggle Bank Customer Complaints
- Tasks
 1. Build a MCP server with one tool that searches the complaints by category.
 2. Run the server locally and confirm it starts without errors.
 3. Connect it to a Python client using the mcp library.
 4. Ask one question that requires the tool, and show the tool call and the answer.

Summary

- ✓ - What MCP is
- ✓ - Why MCP matters
- ✓ - MCP vs API
- ✓ - Host, client, server
- ✓ - Tools, resources, prompts
- ✓ - How the LLM calls a tool
- ✓ - MCP and Agentic RAG
- ✓ - Minimal code demo
- ✓ - Where to run an MCP server

Thank You

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